

Why does the construction give a basis of S ?

Reverse the construction: how can we check n given solutions?

Check whether $\Phi(y_1), \dots, \Phi(y_n)$ are _____.

2. The Wronskian test

For n functions $y_1, \dots, y_n$, define their **Wronskian** by

$$W(t) = \det \begin{pmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ \dot{y}_1(t) & \dot{y}_2(t) & \cdots & \dot{y}_n(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{pmatrix}.$$

At $t = t_0$, its columns are the vectors $\Phi(y_1), \dots, \Phi(y_n)$.

Procedure for checking a fundamental set

- (a) Check that $y_1, \dots, y_n$ are _____.
- (b) Pick an arbitrary $t_0 \in I$.
- (c) The functions form an FSS if and only if _____.

Remark. For solutions of (1), if $W(t_0) \neq 0$ for some $t_0 \in I$, then _____ for every $t \in I$.

Example 11.1 – Checking a fundamental set. Consider

$$\ddot{y} + 2\dot{y} + y = 0.$$

We claim that $y_1(t) = e^{-t}$ and $y_2(t) = te^{-t}$ form an FSS on $I = \mathbb{R}$.

- (a) Check that both functions are solutions of the ODE.

- (b) Choose $t_0 = 0$ and evaluate the Wronskian.

(c) State the conclusion.

Therefore $\{e^{-t}, te^{-t}\}$ is a _____ on $\mathbb{R}$.